

LAB NOTES · ISSUE 01 OF 04

Lab: a Terraform GCP landing zone.

A multi-environment foundation, built as a lab — not a customer deployment.

01

AR

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Swipe →

What it provisions

01

Shared VPC

02

Private GKE

03

IAM and service accounts

04

Cloud NAT for egress

05

Artifact Registry

06

Observability and security controls

How a change flows

1 Bootstrap state bucket

2 validate

3 plan per environment

4 Review in CI

5 apply

Remote state is bootstrapped once by a script, then every environment has its own backend.

Design decisions

01 Environments as folders

Each with its own backend and variables.

02 Modules for shared pieces

Artifact Registry and networking are reused.

03 Private nodes + NAT

No public node IPs; egress is controlled.

04 CI on every PR

fmt, validate and plan before review.

Run it locally

BASH

```
git clone https://github.com/iarsingh/terraform-gcp-platform
cd terraform-gcp-platform

./scripts/bootstrap-state-bucket.sh
./scripts/validate.sh
./scripts/plan-all.sh
```

Plan-only by default. Apply belongs in a pipeline with approval.

What I would add for production

☒ Policy checks on the plan JSON

☒ Nightly drift detection

☒ Org policy constraints

☒ Budget alerts per project

☒ Workload identity everywhere



KEY TAKEAWAY

A lab is where you practise the guardrails, not just the resources.

- Shared VPC, private GKE, NAT, IAM
- Per-environment state
- Plan in CI, apply with approval

YOUR TURN

What is the first module you would add to a landing zone like this?

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